

PURPOSE

Elastic bending stress verification for a simply supported rectangular timber beam under uniform load.

1 Input Parameters

Symbol	Value	Unit	Description
b	200	mm	Section width
h	400	mm	Section depth
L	5000	mm	Span
q	4.5	kN/m	Distributed load (design)
f_m	24	MPa	Design bending strength

2 Calculation

Maximum bending moment

$$\begin{aligned}M_{max} &= \frac{q \cdot \left(\frac{L}{1000}\right)^2}{8} \\&= \frac{4.5 \cdot \left(\frac{5000}{1000}\right)^2}{8} \\&= \mathbf{14.06 \text{ kNm}}\end{aligned}$$

Elastic section modulus

$$\begin{aligned}W &= \frac{b \cdot h^2}{6} \\&= \frac{200 \cdot 400^2}{6} \\&= \mathbf{5.333e + 06 \text{ mm}^3}\end{aligned}$$

Maximum bending stress

$$\begin{aligned}\sigma_m &= M_{max} \cdot 1000000 \cdot \frac{1}{W} \\&= 14.06 \cdot 1000000 \cdot \frac{1}{5.333e + 06} \\&= \mathbf{2.637 \text{ MPa}}\end{aligned}$$

3 Verification

OK

$$\sigma_m = 2.637 \text{ MPa} \leq f_m = 24 \text{ MPa}$$